

Final Paper

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Introduction

(*JUNE*) - Establish context for study: - Relevance to NCOS - Relevance to other ecosystems - Synthesize main findings from other papers (connect ideas, don't just list) - Articulate main question(s) for the study

Methods

(*JUNE*) - Describe temporal and spatial coverage of dataset - Reference monitoring report or NCOS-related publications on data collection - Describe frequency of data collection - Describe statistical analysis or descriptive statistics (linked to your question) ->

Results

(*NATALIE*) - Discuss findings clearly (patterns, trends, etc.) - Connect results to your research questions and figures - Present figures in an order that tells a clear ecological story

Discussion

Connecting Results to NCOS Reports/Publications

The results align with findings from Stiles & Kracha, whose poster on a restored California wetland documented shifts in invertebrate assemblages as the system matured over three years (Stiles and Kracha 2023). Given that NCOS restoration began in 2017 and sampling commenced in 2018, the data collected likely reflect an ecosystem still establishing ecological equilibrium. The winter abundance spike observed is consistent with Stiles & Kracha's finding that seasonal rainfall reduces salinity stress, driving fluctuations in invertebrate community composition (Stiles and Kracha 2023).

Connecting Results to Other Ecosystems

The temperature-abundance relationship observed at NCOS finds support in studies conducted across comparable wetland systems. Marchetti et al. found that invertebrate abundance, richness, and diversity increased with wetland age across restored Sacramento Valley wetlands, with Ostracoda — one of the three taxa monitored in this study — comprising over 20% of all samples (Marchetti et al. 2010). Given that NCOS is a relatively young restoration site, the abundance patterns documented here likely reflect an early stage of that same successional trajectory. Suren et al. similarly demonstrated that invertebrate communities recolonized successfully following hydrological restoration with minimal intervention, suggesting that restored wetlands across geographic contexts follow comparable recovery dynamics (Suren et al. 2011).

Challenges in Data Collection

Obtaining representative water samples was complicated by particulate matter — including mud, sticks, and organic debris — that required filtration and likely introduced variability in usable sample volume across collection events. Some sampling locations, such as areas beneath bridges or in seasonally waterlogged margins, posed physical access challenges that may have limited consistency across seasons.

Other Variables That Would Influence Observed Patterns

Salinity is the most directly relevant confounding variable, as winter rainfall reduces salinity independently of temperature, meaning some of the abundance patterns observed may partly reflect salinity dynamics rather than thermal effects alone; salinity is typically measured using a conductivity meter or refractometer at the time of collection. Dissolved oxygen (DO) is an additional variable of interest, given that warmer water holds less oxygen and could suppress invertebrate abundance at elevated temperatures independently of direct thermal stress; DO is measured via probe concurrent with sampling. Local vegetation cover and canopy shading may also introduce spatial variability, as shaded microsites can register cooler temperatures independent of seasonal conditions.

Temporal Changes and Dataset Limitations

Seasonal sampling captures broad temporal trends but cannot account for finer-scale environmental variation; a single sample per season represents a snapshot that may not reflect conditions experienced by invertebrate populations over preceding weeks. Since invertebrate community structure responds to accumulated conditions rather than instantaneous measurements, short-term fluctuations between sampling events go undetected. Without continuously deployed temperature loggers at each of the 12 Devereux Slough sites, it remains difficult

to assess whether recorded temperatures are representative of the broader inter-sampling period, and multi-year data would be needed to distinguish restoration-driven trajectories from recurring seasonal cycles.

References

- Marchetti, Michael P., Mariah Garr, and Adam N. H. Smith. 2010. "Evaluating Wetland Restoration Success Using Aquatic Macroinvertebrate Assemblages in the Sacramento Valley, California." *Restoration Ecology* 18: 457–66. <https://doi.org/10.1111/j.1526-100X.2008.00468.x>.
- Stiles, L., and C. Kracha. 2023. *Assessing Aquatic Invertebrate Establishment in a Restored Wetland over Time*. UC Santa Barbara: Cheadle Center for Biodiversity; Ecological Restoration. <https://escholarship.org/uc/item/00t2k2t7>.
- Suren, Alastair M., Paul Lambert, and Brian K. Sorrell. 2011. "The Impact of Hydrological Restoration on Benthic Aquatic Invertebrate Communities in a New Zealand Wetland." *Restoration Ecology* 19: 747–57. <https://doi.org/10.1111/j.1526-100X.2010.00723.x>.